# Markdown - Simple Cheat Sheet

The basic syntax you need on your first day

Writing Toolkit

2026

## Start here

Markdown is **normal text with a few simple symbols for formatting**.

A Markdown file usually ends in:

`.md`

For example:

`notes.md`

You can write Markdown in **VS Code, GitHub, HackMD, StackEdit, Obsidian, Jupyter, or Quarto**.

> **Beginner rule:** type first, preview second. You do not need to memorize everything.

A good beginner workflow in VS Code is:

```
left: what you type        right: rendered preview
```

Open preview with:

```
macOS:          Shift + Command + V
Windows/Linux: Ctrl + Shift + V
```

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## 1. Headings

### Type this

```
# Main title
## Section
### Smaller section
```

### It looks like this

# Main title

## Section

### Smaller section

**Simple rule:** more `#` signs = smaller heading.

Always put a space after `#`.

```
## Results     good
##Results      avoid
```

---

## 2. Paragraphs

### Type this

```
This is the first paragraph.

This is the second paragraph.
```

### It looks like this

This is the first paragraph.

This is the second paragraph.

**Simple rule:** leave one blank line between paragraphs.

---

## 3. Bold and italic

### Type this

```
**important result**
*paper title or emphasis*
***very important***
```

### It looks like this

**important result**

*paper title or emphasis*

***very important***

A useful research-note pattern is:

```
**Finding:** The estimate is stable.

**Question:** Why does the result change?
```

```
**TODO:** Check Table 3.
```

---

## 4. Bullet and numbered lists

### Bullets - type this

```
- Read the paper
- Check the appendix
- Reproduce Figure 2
```

### It looks like this

- Read the paper
- Check the appendix
- Reproduce Figure 2

### Numbered list - type this

```
1. Download the data
2. Clean the data
3. Run the analysis
```

### It looks like this

1. Download the data
2. Clean the data
3. Run the analysis

---

## 5. Checkboxes

Checkboxes are useful for research tasks.

### Type this

```
- [x] Read the introduction
- [ ] Check the data section
- [ ] Prepare seminar questions
```

### It looks like this

- ☒ Read the introduction
- ☐ Check the data section
- ☐ Prepare seminar questions

  Checkboxes work especially well on GitHub and many Markdown editors.

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## 6. Links

The pattern is:

```
[text people see](web address)
```

### Type this

```
[Google Scholar](https://scholar.google.com/)
```

### It looks like this

Google Scholar

You can also link to another file in the same project:

```
[Open my notes](notes.md)
```

**Simple rule:** square brackets contain the label; parentheses contain the destination.

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## 7. Images

An image looks like a link with `!` at the beginning.

```
![description](path-to-image)
```

Suppose your folder is:

```
project/
  notes.md
  figures/
    result.png
```

Type:

```
![Main result](figures/result.png)
```

**Simple rule:** use a path relative to the Markdown file. Avoid paths such as `/Users/name/Desktop/...`.

If an image does not appear, check the filename and folder path first.

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## 8. Inline code

Use one backtick when you want to show a filename, variable name, command, or code literally.

### Type this

```
Open `analysis.R` and check the variable `income`.
```

### It looks like this

Open `analysis.R` and check the variable `income`.

This is also useful for showing a LaTeX command such as `\qquad` without running it.

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## 9. Code blocks

Use three backticks before and after a larger block of code.

### Type this

````
```r
model <- lm(y ~ x, data = df)
summary(model)
```
````

### It looks like this

```r
model <- lm(y ~ x, data = df)
summary(model)
```

You may replace `r` with `python`, `stata`, `bash`, or another language name.

**Simple rule:** every opening set of three backticks needs a closing set.

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## 10. A small table

Tables are optional. Learn them only when you need them.

### Type this

```
| Variable | Mean |
|---|---:|
| Income | 52.4 |
| Age | 34.1 |
```

### It looks like this

| Variable | Mean |
|---|---:|
| Income | 52.4 |
| Age | 34.1 |

The second row tells Markdown that this is a table.

For complicated publication tables, use LaTeX, Quarto, or generated output instead of forcing everything into Markdown.

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## 11. Simple math - optional

Some Markdown tools support LaTeX-style math.

### Inline math

Type:

```
The parameter is $\beta$.
```

Rendered idea:

The parameter is $\beta$.

### Display math

Type:

```
$$
y_i = \alpha + \beta x_i + \varepsilon_i
$$
```

Rendered idea:

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

Inside the math, `\alpha`, `\beta`, and `\varepsilon` are LaTeX math commands.

If you see the dollar signs and backslashes literally, your current Markdown viewer may not support math in that context.

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## 12. Five beginner mistakes

### No space after a heading marker

```
##Results      wrong
## Results     better
```

### No blank line between paragraphs

Use a blank line when you want a new paragraph.

### Missing closing code fence

Every opening triple-backtick fence needs a closing fence.

### Image path points to your own computer

Prefer:

```
![Result](figures/result.png)
```

instead of an absolute path from your desktop.

### Assuming every Markdown viewer is identical

Basic syntax is very portable, but math, citations, footnotes, and some other features vary by tool.

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## Quick lookup

| I want to… | Type |
|---|---|
| Heading | `# Heading` |
| Smaller heading | `## Heading` |
| Bold | `**text**` |
| Italic | `*text*` |
| Bullet | `- item` |
| Numbered item | `1. item` |
| Checkbox | `- [ ] task` |
| Link | `[name](https://...)` |
| Image | `![description](image.png)` |
| Inline code | `` `code` `` |
| Code block | three backticks before and after |
| Small table | cells separated by vertical bars, plus a separator row |
| Inline math | `$...$` |
| Display math | `$$...$$` |

## A complete first note

Copy this into a file called `paper_notes.md`:

```
# Paper Notes

**Paper:** Example Paper

## Main question

What question does the paper answer?

## Main finding

The main result is ...

## My questions

- Why is this sample used?
- What is the main mechanism?
- What should I check in the appendix?

## Tasks

- [ ] Read the data section
```

```
- [ ] Check Table 2
- [ ] Write a one-paragraph summary
```

That is already a useful Markdown document. Learn the advanced sheet only when you actually need more.